1. What are the benefits of using IoT in Oil and Gas Industries?

a. Decreased production efficiency
b. Save cost and time
c. Deteriorate asset maintenance
d. None of the above

Answer: (b)

Justification: Please refer to Slide 9 of Lecture 1 of Week 12

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2. Which of the following statement does not represents the advantage of using IoT in the chemical industry?

a. Predictive maintenance
b. Reduce energy expenses
c. Maximize supply chain risks
d. Improve logistics

Answer: (c)

Justification: Please refer to Slide 11 of Lecture 1 of Week 12

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3. What is/are the advantages of using IoT in the Pharmaceutical industry?

a. Real-time monitoring
b. Inability to control the areas remotely
c. Increased production cost and wastage
d. All of the above

Answer: (a)

Justification: Please refer to Slide 18 of Lecture 1 of Week 12

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4. Which of the following can be performed through predictive maintenance?

a. Address real-time issues
b. Reduce equipment breakdown
c. Efficient and effective maintenance
d. All of the above

Answer: (d)

Justification: Please refer to Slide 12 of Lecture 1 of Week 12

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5. State true or false — Through the application of IoT in the Pharmaceutical industry, it is possible to examine drugs and detect adverse drug reactions.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 19 of Lecture 1 of Week 12

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6. Which of the following is true about the features of second-generation UAVs?

a. Static design, fixing camera mount, still photography, video recording, and manual steering control
b. Transformable designs, 1080 HD video or higher value instrumentation, three-axis gimbals
c. Improved safety and regulatory, platform and payload adaptability, automated safety modes
d. Enhanced intelligent piloting models and full autonomy, full airspace awareness, auto action

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 2 of Week 12

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7. What is/are the application field(s) of UAVs?

a. Agriculture
b. Mining
c. Forestry
d. All of the above

Answer: (d)

Justification: Please refer to Slide 5 of Lecture 2 of Week 12

8. Which of the following does not represents an application of UAV in Agriculture?

a. Crop health monitoring
b. Monitoring soil quality, counting plants, identifying plot size
c. Inspect construction site infrastructure and construct roadways
d. Increase effective yields

Answer: (c)

Justification: Please refer to Slides 6,7 and 8 of Lecture 2 of Week 12

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9. State true or false — Precision forestry and canopy mapping cannot be performed through the application of UAV in forestry.

a. True
b. False

Answer: (b)

Justification: Please refer to Slides 18 of Lecture 2 of Week 12

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10. Which of the following does not represents the application of UAVs for energy management?

a. Optimized use of seed, fertilizer, and water
b. Eliminate the need to get close to the dangerous wires
c. Inspections without climbing power poles
d. None of the above

Answer: (a)

Justification: Please refer to Slide 10 of Lecture 2 of Week 12

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11. State true or false — One of the applications of UAVs in telecommunication is to perform tower inspection.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 13 of Lecture 2 of Week 12

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12. Match the following.

Sector	Application of UAVs
1. Mining	i. Scan a large number of items available in a warehouse
2. Healthcare	ii. Regular surface survey for optimized blast design
3. Forestry	iii. Delivery of medicines, vaccines, and snake bite serum
4. Warehousing	iv. Display information about the forest species

a. 1-iii, 2-iv, 3-i, 4-ii
b. 1-ii, 2-iii, 3-iv, 4-i
c. 1-i, 2-ii, 3-iii, 4-iv
d. 1-iv, 2-i, 3-ii, 4-iii

Answer: (b)

Justification: Please refer to Slides 11,12,17 and 18 of Lecture 2 of Week 12

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13. What is/are the application(s) of UAVs in oil and gas industry?
a. Reducing manpower requirements and increasing safety by eliminating the need for industrial mountaineering
b. Monitoring pipelines for detecting leakage of oil and gas pipelines
c. Collecting videos and thermal imagery of oil and gas fields
d. All of the above

Answer: (d)

Justification: Please refer to Slide 16 of Lecture 2 of Week 12

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14. State true or false - UAV-based light displays are generally more expensive than traditional firework displays and cannot be reused.

a. True
b. False

Answer: (b)

Justification: Please refer to Slide 20 of Lecture 2 of Week 12

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15. Which of the following is/are true about the necessity of case studies?

a. The case study investigates real-life phenomena through a detailed analysis of related events.
b. The case study generally selects a vast geographical area or a large number of individuals as the subject matter.
c. Both (a) and (b)
d. Neither (a) nor (b)

Answer: (a)

Justification: Please refer to Slide 3 of Lecture 3 of Week 12
